| | SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P) |
|---|---|
| **Department** | Mechanical Department |
| **Branch** | All Branches |
| **Year & Sem** | B.Tech. First Year (I - Semester) |
| **Session** | Batch 2026-27 onwards |

| Category of Course | Course Title | Course Code | Credits - 3 | | | Theory Paper |
|---|---|---|---|---|---|---|
| | | | L | T | P | |
| ESC | Fundamentals of Mechanical Engineering | ME26101 | 3 | 0 | 0 | Max.Marks - 70<br>Min.Marks - 22<br>Duration - 3 Hrs. |

| Course Code | Course Title | Maximum Marks Allotted | | | | | | Total Marks |
|---|---|---|---|---|---|---|---|---|
| | | Theory Paper | | | Practical | | | |
| | | End Sem. | Mid Sem. MST | Quiz/ Assignment | End Sem. | Lab Work | Assign/Quiz /Term paper | |
| ME26101 | Fundamentals of Mechanical Engineering | 70 | 20 | 10 | -- | -- | -- | 100 |

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| **Course Objectives** | The objectives of this course is to:<br>• Impart fundamentals of Mechanical Engineering to the students<br>• Understand the basics of thermal, materials, manufacturing and renewable energy<br>• Provide knowledge of relevant interdisciplinary engineering application<br>• Introduce the latest trends such as smart materials, additive manufacturing, EV etc. |
| **Prerequisite Knowledge** | Intermediate Level Physics |
| **Course Description** | This course provides the basic knowledge of thermodynamics, fluid mechanics, materials, manufacturing processes and renewable energy systems. The course focuses on both theoretical knowledge of mechanical engineering as well as real world engineering applications. |
| **Course Outcomes** | Upon completion of the course, student will be able to<br>1. Understand engineering materials, mechanical behavior, testing methods, and smart materials<br>2. Understand basic concepts of thermodynamics and working principles thermodynamic systems<br>3. Gain knowledge of conventional and advanced fabrication technologies.<br>4. Understand renewable energy systems, electric vehicles, and energy storing Devices<br>5. Learn the fundamentals of fluid mechanics and heat transfer |

**CO-PO Mapping**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | | | 2 | 1 | | | | | | 1 | | | |
| CO2 | 3 | 1 | | | 1 | 1 | | | | | | 1 | | | |
| CO3 | 3 | 1 | | | 3 | 1 | 1 | | | | | 2 | | | |
| CO4 | 3 | 2 | | | 2 | 2 | 3 | | | | | 3 | | | |
| CO5 | 3 | 1 | | | 2 | 1 | 2 | | | | | 1 | | | |

**(0 = Not correlated; 1 = weakly correlated; 2 = Moderately correlated; 3 = strongly correlated)**

## Syllabus

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|---|---|---|
| Unit - I | Engineering Materials: Metals and alloys, Steels, Ceramics, Polymers, Composite materials; Smart Materials: Shape memory alloys, Piezoelectric materials, Nanomaterials; mechanical properties, testing and applications of engineering materials | CO1 |
| Unit - II | Thermal engineering and its application: Properties of steam and their calculation from steam table and mollier chart. Introduction of IC Engines: Classifications, Otto cycle, diesel cycle, Thermal efficiency, modes of heat transfer; Refrigeration and air Conditioning: Refrigeration, Refrigerants and their properties, Vapor compression refrigeration system (basic idea), Coefficient of Performance (COP) | CO2 |
| Unit – III | Fabrication Processes: Brief introduction of Casting, Forming, Joining (Welding, brazing and soldering); Machining Processes: Lathe, milling, drilling and grinding machines, Machine tools and cutting tools, Introduction of nontraditional machining, NC and CNC machine, 3D printing | CO3 |
| Unit - IV | Introduction to Renewable Energy Sources: Introduction to renewable energy, Solar photovoltaic systems, Solar thermal systems, Wind Energy, Tidal Energy, Geothermal Energy, Electric Vehicles (EV), Hybrid EVs; Energy Storing Devices: Batteries (Lead acid, Lithium ion based), Battery Thermal Management System, Hydrogen Fuel Cells (Introduction) | CO4 |
| Unit – V | Fundamentals of Fluid Mechanic: Fluid Properties, Pressure and its measurement, Pascal's law and Bernoulli's theorem, Laminar and turbulent flow, Reynolds Number, Centrifugal pump, Hydraulic turbines (brief introduction) | CO5 |

**Textbooks:** 1. Basic Mechanical Engineering by R.K. Rajput
2. Engineering Thermodynamics by P.K. Nag
3. Materials Science and Engineering by William D. Callister
4. Manufacturing Engineering and Technology by Serope Kalpakjian

**Reference Books:** 1. Elements of Mechanical Engineering by S. Trymbaka Murthy
2. Additive Manufacturing Technologies by Ian Gibson
3. Electric and Hybrid Vehicles by Iqbal Husain

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| **Department** | Mechanical Engineering Department |
| **Branch** | Mechanical Engineering |
| **Year & Sem** | B.Tech. First Year (I - Semester) |
| **Session** | Batch 2026-27 onwards |

| Category of Course | Course Title | Course Code | L | T | P | Credits - 1 |
|---|---|---|---|---|---|---|
| DLC | Fundamentals of Mechanical Engineering | ME26101P | 0 | 0 | 2 | |

| Course Code | Course Title | Maximum Marks Allotted | | | | | Total Marks |
|---|---|---|---|---|---|---|---|
| | | Theory Paper | | | Practical | | |
| | | End Sem. | Mid Sem. MST | Quiz/ Assignment | End Sem. | Lab Work | |
| ME26101P | Fundamentals of Mechanical Engineering | -- | -- | -- | 30 | 20 | 50 |

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| **Course Objectives** | 1. To provide practical exposure to the fundamental concepts and principles of mechanical engineering through laboratory experiments and demonstrations so as to bridge the gap between theoretical knowledge and practical applications in the field of mechanical engineering.<br>2. To develop understanding of the working and performance characteristics of basic mechanical systems such as boilers, internal combustion engines, refrigeration and air-conditioning systems.<br>3. To familiarize students with commonly used measuring instruments, metrology equipment, and material testing techniques used in engineering practice. |
| **Prerequisite Knowledge** | Students should have basic knowledge of thermodynamics, steam generation, and Internal Combustion (I.C.) engine concepts. |
| **Course Description** | The course emphasizes experimental observation, data analysis, report preparation, and the application of theoretical concepts to real engineering practices, thereby developing essential technical and problem-solving skills required for engineering education and industry. |
| **Course Outcomes** | **Upon completion of the course, student will be able to (Please co-relate 05 Co's for 10 practical's)**<br>1. Compare the construction and working principle of different types of steam boilers.<br>2. Develop a critical understanding of the application of different types of fire tube boilers and water tube boiler.<br>3. Interpret the construction and working of different types of Internal Combustion (I.C.) engines.<br>4. Apply the working principle of Micrometer, Slip Gauges, Dial Gauge, Sine-bar and machine tools.<br>5. Understand the working of Universal testing machie and Vapor compression refrigeration systems |

## List of Experiments

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|---|---|---|
| 1. | Analyze the general construction and classification of steam generators. | CO 1 |
| 2. | Identify different types of boiler mountings and accessories on the basis of their working principle and implementation. | CO 2 |
| 3. | Analyze the working principle and application of some specific fire tube and water tube boilers | CO 2 |
| 4. | Classify and compare different types of internal combustion engines. | CO 2 |
| 5. | Analyze the working principle of two stroke petrol and diesel engine through their construction details and property coordinate diagrams. | CO 3 |
| 6. | Analyze the working principle of four stroke petrol and diesel engine through their construction details and property coordinate diagrams. | CO 3 |
| 7. | Linear and Angular measurement using micrometer, slip Gauges, dial Gauge and sine-bar | CO 4 |
| 8. | Demonstration of the working of machine tools: Lathe, Milling, Drilling and Shaper machine | CO 4 |
| 9. | .Develop a critical understanding of the failure of mild steel and cast iron under tension and compression on Universal Testing Machine. | CO 5 |
| 10 | To perform a trial on Vapor Compression Refrigeration Test Rig. | CO 5 |